

Marcelo Gonzales

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Summary

Mechanical and software engineer with experience supporting safety-critical aerospace systems, engineering analysis, integration, verification, and test automation.

Experience

The Boeing Company

Vehicle Management Software Engineer

Houston, TX

May 2023 - July 2026

CST-100 Starliner

- Developed and maintained embedded flight software for a human-rated spacecraft using C in a real-time operating system environment.
- Performed software integration, verification, debugging, and requirements-based testing for safety-critical spacecraft functions.
- Created and maintained test automation suite using Perl, Python and Jenkins.

MQ-25

- Developed and maintained C# test software for requirements-based testing and verification of MQ-25 mission-management flight software, supporting flight-readiness activities.

International Space Station

- Developed C and Python software tools supporting International Space Station flight operations and mission-support.
- Created and maintained database utilities used to manage, validate, and retrieve mission-support data.
- Redesigned and refactored legacy test frameworks to increase test automation, improve requirements traceability, and strengthen software verification throughout the development life cycle.

Invocon, Inc.

Software Design Engineer

Conroe, TX

October 2019 - March 2023

- Developed embedded firmware in C using FreeRTOS for a smart-battery system.
- Developed and maintained C# and Python GUIs for real-time monitoring of hardware health and status.
- Built Python library and developed standalone scripts to automate verification across multiple products, including SSI, KIPS, and iPCMe.
- Designed custom cable assemblies and test equipment to support hardware integration, functional testing, and verification of products.

Education

Colorado School of Mines

B.S. in Mechanical Engineering

B.S. in Computer Science

Golden, CO

August 2019

Technical Skills

Mechanical: SolidWorks, 3D modeling, manufacturing drawings, BOMs, cable assemblies, test fixtures, test equipment

Hardware/Test: Integration, functional testing, troubleshooting, requirements-based verification

Software: C, C#, Python, PID control, GUI development, test automation